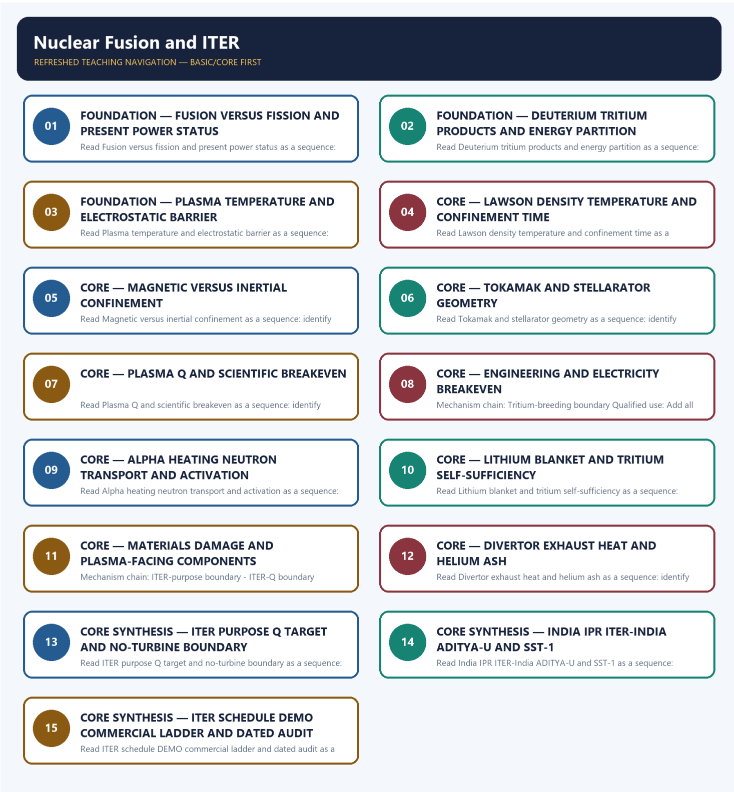
COMPLETE LEARNING SESSION

Nuclear Fusion and ITER - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

Nuclear Fusion and ITER - Learner-v2 Complete Learning Session

> **Authoring-only generation:** 2026-09-03. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-03.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable official UPSC papers were supplementary only for printed routed demands. No answer key, payload capacity, mission result, constellation health, reactor output, isotope value or fusion milestone was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Audited ledgers route the direct 2025 GS-III ITER contribution and future-energy demand here. No other direct PYQ or answer key is manufactured.
- **Live-link boundary:** Substantive ITER text fetched on 2026-09-03 confirms the experimental Q=10 objective, no-electricity boundary, technology-integration purpose and tritium-breeding test role. Schedule, cost and package-status claims remain date-bounded.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-03. Substantive official text is used only for the proposition it supports; title-only, blocked or thin pages are recorded and supply no factual claim.

- https://www.iter.org/fusion-energy/what-will-iter-do - attempted 2026-09-03; substantive ITER text confirmed the experimental Q=10 design objective of 500 MW fusion power from 50 MW injected heating, that ITER will not convert that heat to electricity, and that it will test integrated technologies and tritium-breeding concepts. These claims were not converted into plant-wide or commercial breakeven.

BASIC LEARNING SESSION

DEEP-REVIEW LEARNING CONTRACT

Control	Binding rule for this package
Syllabus boundary	Complete Science and Technology Basic/Core is easy-first and answer-complete before optional Advanced depth.
System boundary	Organisation, platform, subsystem, payload, process, application, institution and outcome remain distinct.
Technical boundary	Physical mechanism, classification, stage, platform, unit, range/capacity and limiting condition are explicit.
Status boundary	Announcement, approval, development, test, deployment, induction, commercial operation and observed outcome remain distinct.
Data boundary	Every mission, launch, target, capacity, budget, range, timeline, rule and regulatory claim keeps source, date, unit and status.
Governance boundary	Funder, developer, operator, authoriser, regulator, procurer, settlement actor and adjudicator are not conflated.
Practice contract	Every solved item has demand decoding, detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement.
PYQ contract	Official wording and keys are asserted only when verified; routed or reconstructed demands remain labelled.
Dual-flow contract	Graphical and ASCII masters independently reconstruct the same complete Basic-to-optional-Advanced evidence spine.
Approval	This immutable successor remains `approved: false` pending explicit approval.

Canonical Basic/Core owner: `upsc-ai-kit\knowledge\Science-and-Technology\basic\05_Nuclear-Fusion-and-ITER.md`
Substantive canonical provenance owner: `upsc-ai-kit\knowledge\Science-and-Technology\learning-sessions\v2\subject-wide-syllabus\science-and-technology-05_Learning-Session.md`
Optional Advanced owner: `upsc-ai-kit\knowledge\Science-and-Technology\advanced\05_Nuclear-Fusion-and-ITER.md`
Official syllabus mapping: `upsc-ai-kit\knowledge\Science-and-Technology\OFFICIAL-UPSC-SYLLABUS-MAPPING.md`

EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- ISRO organisation, launcher stages, payload and orbit claims retain their exact object and mission date.
- NavIC, GAGAN, human-spaceflight, planetary, nuclear, fusion, missile and procurement claims retain category and status.
- Aadhaar/UPI, AI, quantum and semiconductor claims retain actor, layer, metric, maturity and governance boundaries.
- DPDP/cybersecurity, biotechnology and genetic-engineering claims retain legal/regulatory stage and competent institution.
- Targets, budgets, capacities, ranges and timelines are never promoted into achieved outcomes without dated evidence.
- **Current-status note, rechecked 2026-09-04:** volatile claims retain source/date/status; failed official fetches remain explicit and supply no invented fact.

Generation-local live/current sources:

- `https://www.iter.org/fusion-energy/what-will-iter-do - attempted 2026-09-03; substantive ITER text confirmed the experimental Q=10 design objective of 500 MW fusion power from 50 MW injected heating, that ITER will not convert that heat to electricity, and that it will test integrated technologies and tritium-breeding` concepts. These claims were not converted into plant-wide or commercial breakeven.

Topic-routed verified PYQ owners:

- No topic-local official question file is claimed; repository PYQ routing ledgers control wording and status.

Refreshed teaching navigation

Distinct embedded teaching-navigation image. The separate continuous Cārvāka-style flowchart package remains an independent at-a-glance artifact.

SESSION 1 - FOUNDATION - FUSION VERSUS FISSION AND PRESENT POWER STATUS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not merge fusion with fission power.

Technical definition: Read Fusion versus fission and present power status as a sequence: identify Fusion-fission boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Fusion versus fission and present power status as a sequence: identify Fusion-fission boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Fusion versus fission**
- **present power status**
- **Mechanism chain**
- **Qualified use**
- **Nuclear**

How to use them: Frame the answer through FOUNDATION; define Fusion versus fission, connect present power status with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
FUSION VERSUS FISSION AND PRESENT POWER STATUS
01. Fusion-fission boundary
    |
    v
02. DT-reaction boundary
STATUS / CATEGORY FIREWALL -> Do not merge fusion with fission power.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

SIMPLE SYSTEM EXAMPLE

Read Fusion versus fission and present power status as a sequence: identify Fusion-fission boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge fusion with fission power. This keeps Fusion-fission boundary and DT-reaction boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.
- The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

EXAMINER CAUTION

- Do not merge fusion with fission power.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate present fission electricity from experimental fusion.

MINI RECAP

- **Mechanism chain:** Fusion-fission boundary -> DT-reaction boundary
- **Qualified use:** Separate present fission electricity from experimental fusion.

CLOSING RECALL FLOW - FOUNDATION - FUSION VERSUS FISSION AND PRESENT POWER STATUS

```
START / CONCEPT: FOUNDATION - Fusion versus fission and present power status
        |
        v
EXACT TERMS: FOUNDATION · Fusion versus fission · present power status · Mechanism chain ·
Qualified use · Nuclear
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Fusion-fission boundary - DT-reaction boundary Qualified use:
 Separate present fission electricity from experimental fusion.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not merge fusion with fission power.
        |
        v
UPSC TRAP / ANSWER-USE: Nuclear fusion combines light nuclei, while nuclear fission splits heavy
nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.
        |
        v
ANSWER-GRABBING FORMULATION: Read Fusion versus fission and present power status as a sequence:
identify Fusion-fission boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 2 - FOUNDATION - DEUTERIUM TRITIUM PRODUCTS AND ENERGY PARTITION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not quote temperature as proof of energy gain.

Technical definition: Read Deuterium tritium products and energy partition as a sequence: identify Temperature-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Deuterium tritium products and energy partition as a sequence: identify Temperature-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Deuterium tritium products**
- **energy partition**
- **Mechanism chain**
- **Qualified use**
- **Fusion**

How to use them: Frame the answer through FOUNDATION; define Deuterium tritium products, connect energy partition with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
DEUTERIUM TRITIUM PRODUCTS AND ENERGY PARTITION
01. Temperature-plasma boundary
STATUS / CATEGORY FIREWALL -> Do not quote temperature as proof of energy gain.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

SIMPLE SYSTEM EXAMPLE

Read Deuterium tritium products and energy partition as a sequence: identify Temperature-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not quote temperature as proof of energy gain. This keeps Temperature-plasma boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

EXAMINER CAUTION

- Do not quote temperature as proof of energy gain.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Trace energy between alpha particle and neutron.

MINI RECAP

- **Mechanism chain:** Temperature-plasma boundary
- **Qualified use:** Trace energy between alpha particle and neutron.

CLOSING RECALL FLOW - FOUNDATION - DEUTERIUM TRITIUM PRODUCTS AND ENERGY PARTITION

```
START / CONCEPT: FOUNDATION - Deuterium tritium products and energy partition
        |
        v
EXACT TERMS: FOUNDATION · Deuterium tritium products · energy partition · Mechanism chain ·
Qualified use · Fusion
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Temperature-plasma boundary Qualified use: Trace energy
between alpha particle and neutron.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not quote temperature as proof of energy gain.
        |
        v
UPSC TRAP / ANSWER-USE: Fusion fuel must become an ultra-hot ionised plasma to overcome
electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain
or plant readiness.
        |
        v
ANSWER-GRABBING FORMULATION: Read Deuterium tritium products and energy partition as a sequence:
identify Temperature-plasma boundary, follow the named mechanism to its user or output, and stop at
the last verified status rather than assuming the next stage.
```

SESSION 3 - FOUNDATION - PLASMA TEMPERATURE AND ELECTROSTATIC BARRIER

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Lawson boundary Qualified use: Explain why temperature is necessary but insufficient.

Technical definition: Read Plasma temperature and electrostatic barrier as a sequence: identify Lawson boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

MUST-WRITE KEYWORDS

- **FOUNDATION**
- **Plasma temperature**
- **electrostatic barrier**
- **Mechanism chain**
- **Qualified use**
- **The Lawson**

How to use them: Frame the answer through FOUNDATION; define Plasma temperature, connect electrostatic barrier with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
PLASMA TEMPERATURE AND ELECTROSTATIC BARRIER
01. Lawson boundary
STATUS / CATEGORY FIREWALL -> Do not omit confinement time from fusion conditions.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

SIMPLE SYSTEM EXAMPLE

Read Plasma temperature and electrostatic barrier as a sequence: identify Lawson boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not omit confinement time from fusion conditions. This keeps Lawson boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

EXAMINER CAUTION

- Do not omit confinement time from fusion conditions.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Explain why temperature is necessary but insufficient.

MINI RECAP

- **Mechanism chain:** Lawson boundary
- **Qualified use:** Explain why temperature is necessary but insufficient.

CLOSING RECALL FLOW - FOUNDATION - PLASMA TEMPERATURE AND ELECTROSTATIC BARRIER

```
START / CONCEPT: FOUNDATION - Plasma temperature and electrostatic barrier
        |
        v
EXACT TERMS: FOUNDATION · Plasma temperature · electrostatic barrier · Mechanism chain · Qualified
use · The Lawson
        |
        v
MECHANISM / ARGUMENT: Read Plasma temperature and electrostatic barrier as a sequence: identify
Lawson boundary, follow the named mechanism to its user or output, and stop at the last verified
status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: Mechanism chain: Lawson boundary Qualified use: Explain why temperature is
necessary but insufficient.
        |
        v
UPSC TRAP / ANSWER-USE: Mains: Explain why temperature is necessary but insufficient.
        |
        v
ANSWER-GRABBING FORMULATION: The Lawson triple product links plasma density, temperature and energy-
confinement time; improving one variable alone does not satisfy the fusion condition.
```

SESSION 4 - CORE - LAWSON DENSITY TEMPERATURE AND CONFINEMENT TIME

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not merge magnetic and inertial confinement.

Technical definition: Read Lawson density temperature and confinement time as a sequence: identify Magnetic-inertial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Lawson density temperature and confinement time as a sequence: identify Magnetic-inertial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Lawson density temperature**
- **confinement time**
- **Mechanism chain**
- **Qualified use**
- **Magnetic**
- **Read Lawson**

How to use them: Frame the answer through Lawson density temperature; define confinement time, connect Mechanism chain with Qualified use to explain the mechanism, and use Magnetic for the decisive comparison or qualification.

VISUAL FIRST

```
LAWSON DENSITY TEMPERATURE AND CONFINEMENT TIME
01. Magnetic-inertial boundary
STATUS / CATEGORY FIREWALL -> Do not merge magnetic and inertial confinement.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

SIMPLE SYSTEM EXAMPLE

Read Lawson density temperature and confinement time as a sequence: identify Magnetic-inertial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not merge magnetic and inertial confinement. This keeps Magnetic-inertial boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

EXAMINER CAUTION

- Do not merge magnetic and inertial confinement.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Use the triple product to show simultaneous requirements.

MINI RECAP

- **Mechanism chain:** Magnetic-inertial boundary
- **Qualified use:** Use the triple product to show simultaneous requirements.

CLOSING RECALL FLOW - CORE - LAWSON DENSITY TEMPERATURE AND CONFINEMENT TIME

```
START / CONCEPT: CORE - Lawson density temperature and confinement time
        |
        v
EXACT TERMS: Lawson density temperature · confinement time · Mechanism chain · Qualified use ·
Magnetic · Read Lawson
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Magnetic-inertial boundary Qualified use: Use the triple
product to show simultaneous requirements.
        |
        v
CONSEQUENCE / CONTRAST: Magnetic confinement holds a lower-density plasma with magnetic fields,
while inertial confinement rapidly compresses a small fuel target; the two approaches use different
time and density regimes.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not merge magnetic and inertial confinement.
        |
        v
ANSWER-GRABBING FORMULATION: Read Lawson density temperature and confinement time as a sequence:
identify Magnetic-inertial boundary, follow the named mechanism to its user or output, and stop at
the last verified status rather than assuming the next stage.
```

SESSION 5 - CORE - MAGNETIC VERSUS INERTIAL CONFINEMENT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call every magnetic device a tokamak.

Technical definition: Read Magnetic versus inertial confinement as a sequence: identify Tokamak-stellarator boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Magnetic versus inertial confinement as a sequence: identify Tokamak-stellarator boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Magnetic versus inertial confinement**
- **Mechanism chain**
- **Qualified use**
- **A tokamak**
- **Read Magnetic**
- **Tokamak-stellarator**

How to use them: Frame the answer through Magnetic versus inertial confinement; define Mechanism chain, connect Qualified use with A tokamak to explain the mechanism, and use Read Magnetic for the decisive comparison or qualification.

VISUAL FIRST

```
MAGNETIC VERSUS INERTIAL CONFINEMENT
01. Tokamak-stellarator boundary
STATUS / CATEGORY FIREWALL -> Do not call every magnetic device a tokamak.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

SIMPLE SYSTEM EXAMPLE

Read Magnetic versus inertial confinement as a sequence: identify Tokamak-stellarator boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call every magnetic device a tokamak. This keeps Tokamak-stellarator boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

EXAMINER CAUTION

- Do not call every magnetic device a tokamak.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Compare density, confinement method and pulse duration.

MINI RECAP

- **Mechanism chain:** Tokamak-stellarator boundary
- **Qualified use:** Compare density, confinement method and pulse duration.

CLOSING RECALL FLOW - CORE - MAGNETIC VERSUS INERTIAL CONFINEMENT

```
START / CONCEPT: CORE - Magnetic versus inertial confinement
        |
        v
EXACT TERMS: Magnetic versus inertial confinement · Mechanism chain · Qualified use · A tokamak ·
Read Magnetic · Tokamak-stellarator
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Tokamak-stellarator boundary Qualified use: Compare density,
confinement method and pulse duration.
        |
        v
CONSEQUENCE / CONTRAST: A tokamak is a toroidal magnetic-confinement device using a strong plasma
current, while a stellarator relies on externally shaped coils for steady confinement; neither term
means a commercial power plant.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not call every magnetic device a tokamak.
        |
        v
ANSWER-GRABBING FORMULATION: Read Magnetic versus inertial confinement as a sequence: identify
Tokamak-stellarator boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 6 - CORE - TOKAMAK AND STELLARATOR GEOMETRY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call a tokamak a power plant.

Technical definition: Read Tokamak and stellarator geometry as a sequence: identify Q-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Tokamak and stellarator geometry as a sequence: identify Q-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Tokamak**
- **stellarator geometry**
- **Mechanism chain**
- **Qualified use**
- **Fusion**
- **Engineering**

How to use them: Frame the answer through Tokamak; define stellarator geometry, connect Mechanism chain with Qualified use to explain the mechanism, and use Fusion for the decisive comparison or qualification.

VISUAL FIRST

```
TOKAMAK AND STELLARATOR GEOMETRY
01. Q-plasma boundary
    |
    v
02. Engineering-breakeven boundary
STATUS / CATEGORY FIREWALL -> Do not call a tokamak a power plant.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

SIMPLE SYSTEM EXAMPLE

Read Tokamak and stellarator geometry as a sequence: identify Q-plasma boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call a tokamak a power plant. This keeps Q-plasma boundary and Engineering-breakeven boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

EXAMINER CAUTION

- Do not call a tokamak a power plant.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate device geometry from power-plant status.

MINI RECAP

- **Mechanism chain:** Q-plasma boundary -> Engineering-breakeven boundary
- **Qualified use:** Separate device geometry from power-plant status.

CLOSING RECALL FLOW - CORE - TOKAMAK AND STELLARATOR GEOMETRY

```
START / CONCEPT: CORE - Tokamak and stellarator geometry
        |
        v
EXACT TERMS: Tokamak · stellarator geometry · Mechanism chain · Qualified use · Fusion · Engineering
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Q-plasma boundary - Engineering-breakeven boundary Qualified
use: Separate device geometry from power-plant status.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not call a tokamak a power plant.
        |
        v
UPSC TRAP / ANSWER-USE: Do not call a tokamak a power plant.
        |
        v
ANSWER-GRABBING FORMULATION: Read Tokamak and stellarator geometry as a sequence: identify Q-plasma
boundary, follow the named mechanism to its user or output, and stop at the last verified status
rather than assuming the next stage.
```

SESSION 7 - CORE - PLASMA Q AND SCIENTIFIC BREAKEVEN

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not equate plasma Q with net electricity.

Technical definition: Read Plasma Q and scientific breakeven as a sequence: identify Alpha-neutron boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Plasma Q and scientific breakeven as a sequence: identify Alpha-neutron boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Plasma Q**
- **scientific breakeven**
- **Mechanism chain**
- **Qualified use**
- **Charged**
- **MeV**

How to use them: Frame the answer through Plasma Q; define scientific breakeven, connect Mechanism chain with Qualified use to explain the mechanism, and use Charged for the decisive comparison or qualification.

VISUAL FIRST

```
PLASMA Q AND SCIENTIFIC BREAKEVEN
01. Alpha-neutron boundary
STATUS / CATEGORY FIREWALL -> Do not equate plasma Q with net electricity.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

SIMPLE SYSTEM EXAMPLE

Read Plasma Q and scientific breakeven as a sequence: identify Alpha-neutron boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not equate plasma Q with net electricity. This keeps Alpha-neutron boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

EXAMINER CAUTION

- Do not equate plasma Q with net electricity.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Define the numerator and denominator of plasma Q.

MINI RECAP

- **Mechanism chain:** Alpha-neutron boundary

- **Qualified use:** Define the numerator and denominator of plasma Q.

CLOSING RECALL FLOW - CORE - PLASMA Q AND SCIENTIFIC BREAKEVEN

```
START / CONCEPT: CORE - Plasma Q and scientific breakeven
        |
        v
EXACT TERMS: Plasma Q · scientific breakeven · Mechanism chain · Qualified use · Charged · MeV
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Alpha-neutron boundary Qualified use: Define the numerator
and denominator of plasma Q.
        |
        v
CONSEQUENCE / CONTRAST: Charged alpha particles can remain magnetically confined and heat the
plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or
activate structures.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not equate plasma Q with net electricity.
        |
        v
ANSWER-GRABBING FORMULATION: Read Plasma Q and scientific breakeven as a sequence: identify Alpha-
neutron boundary, follow the named mechanism to its user or output, and stop at the last verified
status rather than assuming the next stage.
```

SESSION 8 - CORE - ENGINEERING AND ELECTRICITY BREAKEVEN

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Engineering and electricity breakeven as a sequence: identify Tritium-breeding boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: Tritium-breeding boundary Qualified use: Add all plant loads before discussing net electricity.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Engineering and electricity breakeven as a sequence: identify Tritium-breeding boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Engineering**
- **electricity breakeven**
- **Mechanism chain**
- **Qualified use**
- **Tritium**
- **D-T**

How to use them: Frame the answer through Engineering; define electricity breakeven, connect Mechanism chain with Qualified use to explain the mechanism, and use Tritium for the decisive comparison or qualification.

VISUAL FIRST

```
ENGINEERING AND ELECTRICITY BREAKEVEN
01. Tritium-breeding boundary
STATUS / CATEGORY FIREWALL -> Do not omit plant recirculating power.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

SIMPLE SYSTEM EXAMPLE

Read Engineering and electricity breakeven as a sequence: identify Tritium-breeding boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not omit plant recirculating power. This keeps Tritium-breeding boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

EXAMINER CAUTION

- Do not omit plant recirculating power.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Add all plant loads before discussing net electricity.

MINI RECAP

- **Mechanism chain:** Tritium-breeding boundary
- **Qualified use:** Add all plant loads before discussing net electricity.

CLOSING RECALL FLOW - CORE - ENGINEERING AND ELECTRICITY BREAKEVEN

```
START / CONCEPT: CORE - Engineering and electricity breakeven
        |
        v
EXACT TERMS: Engineering · electricity breakeven · Mechanism chain · Qualified use · Tritium · D-T
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Tritium-breeding boundary Qualified use: Add all plant loads
before discussing net electricity.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not omit plant recirculating power.
        |
        v
UPSC TRAP / ANSWER-USE: Do not omit plant recirculating power.
        |
        v
ANSWER-GRABBING FORMULATION: Read Engineering and electricity breakeven as a sequence: identify
Tritium-breeding boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
```

SESSION 9 - CORE - ALPHA HEATING NEUTRON TRANSPORT AND ACTIVATION

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Technical definition: Read Alpha heating neutron transport and activation as a sequence: identify Materials boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Alpha heating neutron transport and activation as a sequence: identify Materials boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Alpha heating neutron transport**
- **activation**
- **Mechanism chain**
- **Qualified use**
- **High-energy**
- **Read Alpha**

How to use them: Frame the answer through Alpha heating neutron transport; define activation, connect Mechanism chain with Qualified use to explain the mechanism, and use High-energy for the decisive comparison or qualification.

VISUAL FIRST

```
ALPHA HEATING NEUTRON TRANSPORT AND ACTIVATION
01. Materials boundary
STATUS / CATEGORY FIREWALL -> Do not treat alpha particles and neutrons identically.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

SIMPLE SYSTEM EXAMPLE

Read Alpha heating neutron transport and activation as a sequence: identify Materials boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not treat alpha particles and neutrons identically. This keeps Materials boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

EXAMINER CAUTION

- Do not treat alpha particles and neutrons identically.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Link alpha self-heating to neutron-driven blanket and damage.

MINI RECAP

- **Mechanism chain:** Materials boundary
- **Qualified use:** Link alpha self-heating to neutron-driven blanket and damage.

CLOSING RECALL FLOW - CORE - ALPHA HEATING NEUTRON TRANSPORT AND ACTIVATION

```
START / CONCEPT: CORE - Alpha heating neutron transport and activation
        |
        v
EXACT TERMS: Alpha heating neutron transport · activation · Mechanism chain · Qualified use · High-
energy · Read Alpha
        |
        v
MECHANISM / ARGUMENT: High-energy neutrons cause displacement damage, helium production and
activation, while plasma-facing components face intense heat loads; material survival is separate
from plasma confinement.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not treat alpha particles and neutrons
identically.
        |
        v
UPSC TRAP / ANSWER-USE: Do not treat alpha particles and neutrons identically.
        |
        v
ANSWER-GRABBING FORMULATION: Read Alpha heating neutron transport and activation as a sequence:
identify Materials boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
```

SESSION 10 - CORE - LITHIUM BLANKET AND TRITIUM SELF-SUFFICIENCY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Mechanism chain: Divertor boundary Qualified use: Explain why a breeding blanket is a fuel-system requirement.

Technical definition: Read Lithium blanket and tritium self-sufficiency as a sequence: identify Divertor boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Lithium blanket and tritium self-sufficiency as a sequence: identify Divertor boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Lithium blanket**
- **tritium self-sufficiency**
- **Mechanism chain**
- **Qualified use**
- **Read Lithium**
- **Divertor**

How to use them: Frame the answer through Lithium blanket; define tritium self-sufficiency, connect Mechanism chain with Qualified use to explain the mechanism, and use Read Lithium for the decisive comparison or qualification.

VISUAL FIRST

```
LITHIUM BLANKET AND TRITIUM SELF-SUFFICIENCY
01. Divertor boundary
STATUS / CATEGORY FIREWALL -> Do not assume tritium is freely available.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

SIMPLE SYSTEM EXAMPLE

Read Lithium blanket and tritium self-sufficiency as a sequence: identify Divertor boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not assume tritium is freely available. This keeps Divertor boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

EXAMINER CAUTION

- Do not assume tritium is freely available.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Explain why a breeding blanket is a fuel-system requirement.

MINI RECAP

- **Mechanism chain:** Divertor boundary
- **Qualified use:** Explain why a breeding blanket is a fuel-system requirement.

CLOSING RECALL FLOW - CORE - LITHIUM BLANKET AND TRITIUM SELF-SUFFICIENCY

```
START / CONCEPT: CORE - Lithium blanket and tritium self-sufficiency
        |
        v
EXACT TERMS: Lithium blanket · tritium self-sufficiency · Mechanism chain · Qualified use · Read
Lithium · Divertor
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Divertor boundary Qualified use: Explain why a breeding
blanket is a fuel-system requirement.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not assume tritium is freely available.
        |
        v
UPSC TRAP / ANSWER-USE: A divertor extracts heat, impurities and helium ash from the plasma edge;
successful core confinement does not solve exhaust and component-lifetime constraints.
        |
        v
ANSWER-GRABBING FORMULATION: Read Lithium blanket and tritium self-sufficiency as a sequence:
identify Divertor boundary, follow the named mechanism to its user or output, and stop at the last
verified status rather than assuming the next stage.
```

SESSION 11 - CORE - MATERIALS DAMAGE AND PLASMA-FACING COMPONENTS

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Read Materials damage and plasma-facing components as a sequence: identify ITER-purpose boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

Technical definition: Mechanism chain: ITER-purpose boundary - ITER-Q boundary Qualified use: Treat material lifetime as an independent feasibility threshold.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Materials damage and plasma-facing components as a sequence: identify ITER-purpose boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Materials damage**
- **plasma-facing components**
- **Mechanism chain**
- **Qualified use**
- **ITER**
- **ITER's official design objective**

How to use them: Frame the answer through Materials damage; define plasma-facing components, connect Mechanism chain with Qualified use to explain the mechanism, and use ITER for the decisive comparison or qualification.

VISUAL FIRST

```
MATERIALS DAMAGE AND PLASMA-FACING COMPONENTS
01. ITER-purpose boundary
     |
     v
02. ITER-Q boundary
STATUS / CATEGORY FIREWALL -> Do not call fusion radiation-free or materials-free.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

SIMPLE SYSTEM EXAMPLE

Read Materials damage and plasma-facing components as a sequence: identify ITER-purpose boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call fusion radiation-free or materials-free. This keeps ITER-purpose boundary and ITER-Q boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

EXAMINER CAUTION

- Do not call fusion radiation-free or materials-free.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Treat material lifetime as an independent feasibility threshold.

MINI RECAP

- **Mechanism chain:** ITER-purpose boundary -> ITER-Q boundary
- **Qualified use:** Treat material lifetime as an independent feasibility threshold.

CLOSING RECALL FLOW - CORE - MATERIALS DAMAGE AND PLASMA-FACING COMPONENTS

```
START / CONCEPT: CORE - Materials damage and plasma-facing components
        |
        v
EXACT TERMS: Materials damage · plasma-facing components · Mechanism chain · Qualified use · ITER ·
ITER's official design objective
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: ITER-purpose boundary - ITER-Q boundary Qualified use: Treat
material lifetime as an independent feasibility threshold.
        |
        v
CONSEQUENCE / CONTRAST: ITER's official design objective is 500 MW of fusion power from 50 MW of
injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to
electricity.
        |
        v
UPSC TRAP / ANSWER-USE: ITER is an experimental tokamak under construction to study burning-plasma
conditions and integrated fusion technologies; it has no turbine and is not a commercial
electricity plant.
        |
        v
ANSWER-GRABBING FORMULATION: Read Materials damage and plasma-facing components as a sequence:
identify ITER-purpose boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 12 - CORE - DIVERTOR EXHAUST HEAT AND HELIUM ASH

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not omit divertor and exhaust constraints.

Technical definition: Read Divertor exhaust heat and helium ash as a sequence: identify Membership-contribution boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read Divertor exhaust heat and helium ash as a sequence: identify Membership-contribution boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **Divertor exhaust heat**
- **helium ash**
- **Mechanism chain**
- **Qualified use**
- **ITER**
- **India's**

How to use them: Frame the answer through Divertor exhaust heat; define helium ash, connect Mechanism chain with Qualified use to explain the mechanism, and use ITER for the decisive comparison or qualification.

VISUAL FIRST

```
DIVERTOR EXHAUST HEAT AND HELIUM ASH
01. Membership-contribution boundary
STATUS / CATEGORY FIREWALL -> Do not omit divertor and exhaust constraints.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

SIMPLE SYSTEM EXAMPLE

Read Divertor exhaust heat and helium ash as a sequence: identify Membership-contribution boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not omit divertor and exhaust constraints. This keeps Membership-contribution boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

EXAMINER CAUTION

- Do not omit divertor and exhaust constraints.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Show why exhaust handling limits sustained operation.

MINI RECAP

- **Mechanism chain:** Membership-contribution boundary
- **Qualified use:** Show why exhaust handling limits sustained operation.

CLOSING RECALL FLOW - CORE - DIVERTOR EXHAUST HEAT AND HELIUM ASH

```
START / CONCEPT: CORE - Divertor exhaust heat and helium ash
        |
        v
EXACT TERMS: Divertor exhaust heat · helium ash · Mechanism chain · Qualified use · ITER · India's
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: Membership-contribution boundary Qualified use: Show why
exhaust handling limits sustained operation.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not omit divertor and exhaust constraints.
        |
        v
UPSC TRAP / ANSWER-USE: Do not omit divertor and exhaust constraints.
        |
        v
ANSWER-GRABBING FORMULATION: Read Divertor exhaust heat and helium ash as a sequence: identify
Membership-contribution boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 13 - CORE SYNTHESIS - ITER PURPOSE Q TARGET AND NO-TURBINE BOUNDARY

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: The decisive boundary is Do not call ITER a commercial reactor.

Technical definition: Read ITER purpose Q target and no-turbine boundary as a sequence: identify IPR-ITERIndia boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read ITER purpose Q target and no-turbine boundary as a sequence: identify IPR-ITERIndia boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **ITER purpose Q target**
- **no-turbine boundary**
- **Mechanism chain**
- **Qualified use**
- **IPR**

How to use them: Frame the answer through CORE SYNTHESIS; define ITER purpose Q target, connect no-turbine boundary with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
ITER PURPOSE Q TARGET AND NO-TURBINE BOUNDARY
01. IPR-ITERIndia boundary
STATUS / CATEGORY FIREWALL -> Do not call ITER a commercial reactor.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

SIMPLE SYSTEM EXAMPLE

Read ITER purpose Q target and no-turbine boundary as a sequence: identify IPR-ITERIndia boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not call ITER a commercial reactor. This keeps IPR-ITERIndia boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

EXAMINER CAUTION

- Do not call ITER a commercial reactor.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** State ITER's experimental purpose and no-electricity boundary.

MINI RECAP

- **Mechanism chain:** IPR-ITERIndia boundary
- **Qualified use:** State ITER's experimental purpose and no-electricity boundary.

CLOSING RECALL FLOW - CORE SYNTHESIS - ITER PURPOSE Q TARGET AND NO-TURBINE BOUNDARY

```
START / CONCEPT: CORE SYNTHESIS - ITER purpose Q target and no-turbine boundary
        |
        v
EXACT TERMS: CORE SYNTHESIS · ITER purpose Q target · no-turbine boundary · Mechanism chain ·
Qualified use · IPR
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: IPR-ITERIndia boundary Qualified use: State ITER's
experimental purpose and no-electricity boundary.
        |
        v
CONSEQUENCE / CONTRAST: The decisive boundary is Do not call ITER a commercial reactor.
        |
        v
UPSC TRAP / ANSWER-USE: IPR is a DAE-aided plasma-research institute, while ITER-India is the
domestic agency organised as a project within IPR for procurement and delivery; research institute
and project agency are not synonyms.
        |
        v
ANSWER-GRABBING FORMULATION: Read ITER purpose Q target and no-turbine boundary as a sequence:
identify IPR-ITERIndia boundary, follow the named mechanism to its user or output, and stop at the
last verified status rather than assuming the next stage.
```

SESSION 14 - CORE SYNTHESIS - INDIA IPR ITER-INDIA ADITYA-U AND SST-1

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Technical definition: Read India IPR ITER-India ADITYA-U and SST-1 as a sequence: identify Domestic-device boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **India IPR ITER-India ADITYA-U**
- **SST-1**
- **Mechanism chain**
- **Qualified use**
- **ADITYA-U and SST-1**

How to use them: Frame the answer through CORE SYNTHESIS; define India IPR ITER-India ADITYA-U, connect SST-1 with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
INDIA IPR ITER-INDIA ADITYA-U AND SST-1
01. Domestic-device boundary
    |
    v
02. Schedule-status boundary
STATUS / CATEGORY FIREWALL -> Do not rewrite 500 MW fusion heat as electricity.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

SIMPLE SYSTEM EXAMPLE

Read India IPR ITER-India ADITYA-U and SST-1 as a sequence: identify Domestic-device boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not rewrite 500 MW fusion heat as electricity. This keeps Domestic-device boundary and Schedule-status boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.
- ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

EXAMINER CAUTION

- Do not rewrite 500 MW fusion heat as electricity.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Separate research, domestic agency, devices and in-kind packages.

MINI RECAP

- **Mechanism chain:** Domestic-device boundary -> Schedule-status boundary
- **Qualified use:** Separate research, domestic agency, devices and in-kind packages.

CLOSING RECALL FLOW - CORE SYNTHESIS - INDIA IPR ITER-INDIA ADITYA-U AND SST-1

```
START / CONCEPT: CORE SYNTHESIS - India IPR ITER-India ADITYA-U and SST-1
        |
        v
EXACT TERMS: CORE SYNTHESIS · India IPR ITER-India ADITYA-U · SST-1 · Mechanism chain · Qualified
use · ADITYA-U and SST-1
        |
        v
MECHANISM / ARGUMENT: Read India IPR ITER-India ADITYA-U and SST-1 as a sequence: identify Domestic-
device boundary, follow the named mechanism to its user or output, and stop at the last verified
status rather than assuming the next stage.
        |
        v
CONSEQUENCE / CONTRAST: ITER's re-baselined research, magnetic-energy and deuterium-tritium
milestones are future project targets; construction progress does not convert them into completed
operations.
        |
        v
UPSC TRAP / ANSWER-USE: The decisive boundary is Do not rewrite 500 MW fusion heat as electricity.
        |
        v
ANSWER-GRABBING FORMULATION: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and
```

```
long-pulse research; neither is a power reactor or evidence of commercial fusion generation.
```

SESSION 15 - CORE SYNTHESIS - ITER SCHEDULE DEMO COMMERCIAL LADDER AND DATED AUDIT

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: ITER is designed as an experimental bridge between present fusion machines and later electricity-generating demonstration plants; it is not itself a commercial power plant.

Technical definition: Read ITER schedule DEMO commercial ladder and dated audit as a sequence: identify DEMO-commercial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

> Read ITER schedule DEMO commercial ladder and dated audit as a sequence: identify DEMO-commercial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

MUST-WRITE KEYWORDS

- **CORE SYNTHESIS**
- **ITER schedule DEMO commercial ladder**
- **dated audit**
- **Mechanism chain**
- **Qualified use**
- **Subject**

How to use them: Frame the answer through CORE SYNTHESIS; define ITER schedule DEMO commercial ladder, connect dated audit with Mechanism chain to explain the mechanism, and use Qualified use for the decisive comparison or qualification.

VISUAL FIRST

```
ITER SCHEDULE DEMO COMMERCIAL LADDER AND DATED AUDIT
01. DEMO-commercial boundary
    |
    v
02. Volatile fusion boundary
STATUS / CATEGORY FIREWALL -> Do not invent member shares, cost or contribution values.
```

The rail fixes the technical sequence and the claim boundary before explanation.

CORE EXPLANATION

A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

SIMPLE SYSTEM EXAMPLE

Read ITER schedule DEMO commercial ladder and dated audit as a sequence: identify DEMO-commercial boundary, follow the named mechanism to its user or output, and stop at the last verified status rather than assuming the next stage.

TECHNICAL DISTINCTION

The decisive boundary is Do not invent member shares, cost or contribution values. This keeps DEMO-commercial boundary and Volatile fusion boundary in the correct scientific category, institution and unit of measurement.

NAMED EVIDENCE AND MECHANISM

- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.
- Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

EXAMINER CAUTION

- Do not invent member shares, cost or contribution values.

EXAM LINK

- **Prelims:** Keep institution, vehicle/spacecraft, orbit, service, signal, reactor, fuel, process, unit and status in their exact category.
- **Mains:** Close with scheduled targets, DEMO prerequisites and dated evidence.

MINI RECAP

- **Mechanism chain:** DEMO-commercial boundary -> Volatile fusion boundary
- **Qualified use:** Close with scheduled targets, DEMO prerequisites and dated evidence.

COMPLETE BASIC OWNER EVIDENCE BANK

> **Subject:** Science & Technology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III + Prelims. **Core area:** Fusion basics, tokamaks, confinement methods, ITER and India's domestic fusion research base. **Grounded in:** ITER fusion pages (https://www.iter.org/sci/whatisfusion ; https://www.iter.org/fusion-energy/what-will-iter-do ; https://www.iter.org/fusion-energy/after-iter); ITER India contributions/update pages (https://www.iter.org/node/20687/indias-contributions-iter ; https://www.iter.org/node/20687/updated-baseline-presented ; https://www.iter.org/sites/default/files/media/2025-06/ic-36_press_release_final2.pdf); IPR SST-1 and Aditya-U pages (https://nsd.ipr.res.in/documents/sst-1.html ; https://nsd.ipr.res.in/documents/aditya_u.html); official IPR ecosystem references (https://www.ipr.res.in/); ITER new-baseline release, 8 Jul 2024 (https://www.iter.org/node/20687/new-baseline-prioritize-robust-start-exploitation); ITER mission overview (https://www.iter.org/few-lines); ITER-India contributions page (https://www.iterindia.in/indias-contribution-iter); IPR ADITYA-U and SST-1 pages (https://www.ipr.res.in/rdactivities/Aditya-U1 ; https://nsd.ipr.res.in/documents/sst-1.html) - re-verified 2 Aug 2026. FACT: = source-grounded | ANALYSIS: = analytical inference | CURRENT: = current/dated development. *Companion:* `advanced/05_Nuclear-Fusion-and-ITER.md`.

1. Visual foundation

Contrast	Fusion	Fission
Basic process	Light nuclei combine	Heavy nucleus splits
Present power use	Experimental / research stage	Commercially used in nuclear power plants
Key device in this topic	Tokamak / magnetic confinement	Reactor core in fission plants
Indian syllabus link	ITER, IPR, SST-1, Aditya-U	Three-stage programme, PHWR/FBR (Topic 04)

```
MAGNETIC CONFINEMENT (TOKAMAK)
fuel (deuterium + tritium) -> ultra-hot plasma -> magnetic fields confine plasma
-> fusion reactions -> high-energy neutrons/heat -> future power-extraction concept

INERTIAL CONFINEMENT
tiny fuel pellet -> rapid laser/beam compression -> very short fusion burst
```

2. Essential definitions

Concept	Exam-ready meaning
FACT: **Nuclear fusion**	Combination of light nuclei into a heavier nucleus with release of energy; the process that powers stars.

Concept	Exam-ready meaning
ANALYSIS: **The D-T reaction**	Deuterium (^2H) + Tritium (^3H) -> **Helium-4 + a neutron + 17.6 MeV**. The **14.1 MeV neutron carries ~80% of the energy** out of the plasma (and is what heats the blanket and damages the walls); the **3.5 MeV helium nucleus (alpha) stays in the plasma and heats it** - the basis of "alpha self-heating" or burning plasma.
ANALYSIS: **Why D-T?**	It has the highest reaction cross-section at the lowest attainable temperature (~150 million °C class), so it is the only practically accessible fuel for first-generation reactors. Deuterium is abundant in seawater; **tritium is radioactive with ~12.3-year half-life and does not occur naturally in useful quantities** - hence breeding.
ANALYSIS: **Lawson criterion / triple product**	Net energy requires a sufficient product of **plasma density × temperature × energy-confinement time ($n \cdot T \cdot \tau_E$)**. This single idea explains why fusion is hard: you can raise any one factor easily, but not all three at once.
ANALYSIS: **Fusion gain Q**	Ratio of fusion power produced to external heating power injected. **Q > 1 = scientific breakeven; ITER targets Q ≥ 10; "engineering breakeven" (net electricity after plant consumption) is a much higher and separate bar.**
FACT: **Plasma**	Ionised, extremely hot state of matter in which fusion fuel must be confined for sustained reaction.
FACT: **Tokamak**	Toroidal magnetic-confinement device used to confine plasma for fusion experiments. (A **stellarator** achieves confinement with twisted external coils instead of a large induced plasma current - harder to build, easier to run steadily.)
FACT: **Magnetic confinement**	Fusion approach that uses strong magnetic fields to hold hot plasma away from reactor walls.
FACT: **Inertial confinement**	Fusion approach that compresses tiny fuel pellets rapidly using lasers or beams.
FACT: **ITER**	International experimental fusion project in France intended to demonstrate feasibility, **not** commercial power generation - **ITER has no turbine and will not produce electricity.**
FACT: **SST-1**	Steady State Superconducting Tokamak-1, India's superconducting tokamak at IPR for long-pulse plasma research.
FACT: **Aditya-U**	Upgraded Indian tokamak used for plasma studies and fusion-technology learning relevant to larger programmes.
FACT: **Tritium breeding**	Generating tritium *inside the reactor* by letting fusion neutrons strike a **lithium-bearing blanket** (^6Li + n -> tritium + helium). Without a breeding blanket that produces at least as much tritium as it consumes, a D-T power plant cannot fuel itself.
ANALYSIS: **Neutron damage / activation**	14.1 MeV neutrons displace atoms in structural steel and transmute them, causing swelling, embrittlement and low-level radioactive activation of the *structure*. Fusion produces **no long-lived high-level fission-product waste**, but it is **not "no radioactivity at all"**.
ANALYSIS: **Divertor**	The component that exhausts helium "ash" and heat from the plasma edge. It faces some of the highest sustained heat fluxes in any engineered system, which is why divertor materials are a first-order engineering constraint.

3. Mechanism / how it works

1. Fusion aims to force light nuclei - most commonly deuterium and tritium in laboratory discussion - to combine at extremely high temperature, releasing energy.
2. Because the fuel becomes plasma at such temperatures, it cannot touch ordinary reactor walls directly; it must be confined either magnetically or inertially.
3. A tokamak uses powerful magnetic fields in a toroidal chamber to keep the plasma stable for long enough to study or sustain fusion conditions.
4. Inertial confinement takes a different route: a tiny fuel pellet is compressed and heated rapidly, relying on the pellet's own inertia for a very short fusion event.
5. ITER is designed as an experimental bridge between present fusion machines and later electricity-generating demonstration plants; it is not itself a commercial power plant.

4. Institutions and programmes

- FACT: **ITER Organization:** coordinates the world's flagship experimental tokamak project at Saint-Paul-lez-Durance, France. Its members are **India, the European Union, China, Japan, South Korea, Russia and the United States** - seven parties covering roughly half of humanity.
- FACT: **India as one of the ITER members:** contributes technology, components and scientific participation to the international project, largely **in kind** (manufactured hardware) rather than only in cash.
- FACT: **Institute for Plasma Research (IPR), Gandhinagar:** an **aided institute of the Department of Atomic Energy** and India's principal plasma- and fusion-research institution; it operates ADITYA-U and SST-1.
- FACT: **ITER-India:** a **dedicated project/unit within IPR that acts as India's ITER Domestic Agency** - the legal and procurement interface that places contracts with Indian industry and delivers India's in-kind packages to the ITER Organization. ANALYSIS: **IPR ≠ ITER-India:** IPR is the research institute; ITER-India is the domestic agency housed in it.
- FACT: **India's nine official in-kind procurement packages: cryostat; in-wall shielding; cooling-water system; cryogenic system; ion-cyclotron RF heating; electron-cyclotron RF heating; diagnostic neutral beam; power supplies; and a set of diagnostics.** (The **cryostat**, the largest stainless-steel high-vacuum pressure chamber ever built, is the single most quotable Indian contribution.)
- FACT: **SST-1:** India's superconducting tokamak for longer-pulse, steady-state-relevant plasma research.
- FACT: **Aditya / Aditya-U:** India's earlier and upgraded tokamak platform for plasma physics, diagnostics and operational learning, including deuterium-plasma, divertor and shaped-plasma experiments.
- ANALYSIS: Neither SST-1 nor ADITYA-U is a power reactor: they are physics and technology platforms, and IPR does not describe them as producing energy for the grid.

5. Indian applications, examples and limitations

- FACT: Fusion research builds capability in superconducting magnets, plasma diagnostics, cryogenics, control systems, vacuum engineering and high-end materials.
- FACT: India's participation in ITER gives industrial and scientific exposure far beyond textbooks, linking domestic research to a global frontier project.
- FACT: SST-1 and Aditya-U give Indian laboratories hands-on plasma-research experience rather than leaving fusion as a purely foreign subject.
- ANALYSIS: Limitation: fusion remains a research-stage technology; no operational commercial fusion power plant exists anywhere today.
- ANALYSIS: Limitation: maintaining stable, high-temperature plasma without damaging materials is an enormous scientific and engineering challenge.
- ANALYSIS: Limitation: even if experimental milestones succeed, scaling to economical grid electricity requires later demonstration plants, supply chains, tritium handling and regulatory learning.
- ANALYSIS: Limitation: public debate often mistakes single lab breakthroughs for ready-to-deploy electricity generation; UPSC answers should resist that hype.

6. Must-Know Facts for Prelims

- FACT: Fusion is different from fission: fusion combines light nuclei, while fission splits heavy nuclei.
- FACT: A tokamak is a magnetic-confinement device; ITER follows the magnetic-confinement route, not inertial confinement.
- FACT: ITER is an experimental project intended to demonstrate feasibility and bridge the gap toward later power-producing systems; it is not a commercial plant.
- FACT: India is one of the ITER members and contributes components, expertise and industrial capability to the project.
- FACT: IPR’s Aditya began India’s tokamak journey and its upgraded form Aditya-U continues fusion-relevant research.
- FACT: SST-1 is India’s steady-state superconducting tokamak meant to study longer-pulse plasma operation.
- FACT: Fusion research may eventually support clean energy, but current fusion systems remain at the experimental/research stage.
- FACT: No answer should claim that an operational fusion power plant already exists anywhere in the world.
- FACT: **ITER's design target is Q ≥ 10 - 500 MW of fusion power from 50 MW of injected heating power.** It is a *thermal* power figure; ITER has no electricity-generating equipment.
- FACT: **ITER's 2024 "new baseline" (adopted July 2024)** replaced the symbolic "First Plasma" milestone with **Start of Research Operations in 2034**, full magnetic energy in **2036**, and **deuterium-tritium operation in 2039**. Quote these as *scheduled targets of an under-construction project.*
- FACT: **India's ITER contribution is largely in kind**, across nine packages: cryostat, in-wall shielding, cooling-water system, cryogenic system, ion-cyclotron RF heating, electron-cyclotron RF heating, diagnostic neutral beam, power supplies and diagnostics.
- FACT: **ITER-India is India's Domestic Agency and is a project within IPR**; IPR itself is a DAE-aided plasma-research institute. Do not use the two names interchangeably.
- FACT: **Tritium must be bred in a lithium blanket**; deuterium is abundant in water, but tritium is not naturally available in useful quantity. Fuel self-sufficiency (tritium breeding ratio > 1) is a defining engineering requirement of any future D-T power plant.
- FACT: **Fusion waste profile:** no long-lived fission-product high-level waste and no runaway chain reaction (loss of confinement simply stops the reaction), **but** neutron activation makes structural components radioactive, and tritium is itself a radioactive gas requiring containment.

7. UPSC traps

- WRONG: Fusion and fission are both just “nuclear power” and can be discussed together. -> Fusion is a separate research-stage process; current commercial nuclear plants use fission.
- WRONG: ITER is a power plant that will soon feed electricity to the grid. -> ITER is an experimental facility for demonstrating feasibility, not a commercial electricity plant; it has no turbine.
- WRONG: Tokamak means any fusion approach. -> A tokamak is one magnetic-confinement device; inertial confinement and stellarators are different approaches.
- WRONG: A successful experimental shot means fusion has become commercially viable. -> Experimental success does not automatically solve materials, economics, tritium and grid-scale engineering issues.
- WRONG: India’s fusion research is only through foreign participation. -> India also has domestic tokamak research through IPR, SST-1 and Aditya-U.
- WRONG: Topic 05 can replace Topic 04. -> Topic 05 is about fusion research; Topic 04 is about today’s fission-based nuclear power system.
- WRONG: "Fusion is completely radiation-free and waste-free." -> It avoids long-lived high-level fission waste and meltdown risk, but neutron **activation** of structures and **tritium** handling are real radiological issues.
- WRONG: "Q = 10 means the plant produces ten times the electricity it consumes." -> Q compares **fusion power to injected plasma heating power only**; it ignores magnets, cryoplant, pumps and conversion losses. Engineering breakeven is a far higher bar.

- WRONG: "IPR and ITER-India are the same body." -> ITER-India is the Domestic Agency housed within IPR; the research institute and the procurement agency have different functions.
- WRONG: Deuterium and tritium are equally available. -> Deuterium is abundant in seawater; tritium is radioactive, short-lived and must be bred.

8. CURRENT: Current anchor

- CURRENT: **08 Jul 2024 | ITER new baseline - adopted.** ITER replaced the "First Plasma" framing with **Start of Research Operations in 2034**, full magnetic energy in **2036** and **D-T operation in 2039**, while reaffirming the **Q ≥ 10 / 500 MW from 50 MW** mission target. **Status: under construction, schedule re-baselined.**
- CURRENT: **17 Feb 2025 | India's contributions to ITER - highlighted.** ITER noted the scale of India's industrial and personnel contribution, including the cryostat as the largest of India's procurement packages.
- CURRENT: **19 Jun 2025 | ITER Council Meeting 36 - under-construction project with strong execution.** The Council welcomed ahead-of-schedule sector-module installation and completion of all major magnets while reiterating that ITER remains an experimental facility under construction.
- CURRENT: **Domestic platforms (verified position, Aug 2026):** IPR describes **ADITYA-U** as an experimental tokamak running deuterium-plasma, divertor and shaped-plasma campaigns, and **SST-1** as a superconducting tokamak for steady-state plasma physics. **Status: research platforms - neither is described as a power-producing reactor.**
- CURRENT: **Current status message from official ITER framing:** after ITER, later demonstration plants (DEMO-class) are still the next step; therefore fusion should still be written as experimental/research-stage, not deployed power generation.

ANALYSIS: **Currentness note:** The dated statuses above are accurate to the cited source date (latest re-verification 2 Aug 2026); verify later updates before exam use.

10. Mains framework / angles

- ANALYSIS: A good answer first draws the fusion-fission line sharply, then explains why plasma confinement is the central challenge.
- ANALYSIS: Use ITER as the anchor for global experimental fusion and IPR/SST-1/Aditya-U as the Indian domestic capability base.
- ANALYSIS: Show why fusion excites policymakers - low-carbon potential, abundant fuel logic, no chain-reaction meltdown narrative - but balance this with the research-stage reality.
- ANALYSIS: End with a realistic conclusion: India benefits today mainly through science, engineering and industrial learning, not through immediate fusion electricity.

> **Answer thesis:** Fusion is a scientifically powerful but still experimental energy frontier in which the central question is controlled plasma confinement, not immediate grid supply; India's value lies both in being an ITER member and in sustaining domestic tokamak capability through IPR, SST-1 and Aditya-U while clearly distinguishing fusion from the fission-based nuclear power system.

11. Probable questions

- ANALYSIS: **Prelims (practice):** Which one of the following correctly distinguishes magnetic confinement fusion, inertial confinement fusion and present-day fission power plants?
- ANALYSIS: **Mains (10 marks, practice):** Why must ITER be described as an experimental fusion facility rather than as an operational power plant? Answer in 150 words.
- ANALYSIS: **Mains (15 marks, practice):** Discuss the technological significance of India's participation in ITER and its domestic fusion research ecosystem, while explaining why fusion remains a long-term rather than immediate energy solution.

12. Study links

- FACT: Advanced companion: `advanced/05_Nuclear-Fusion-and-ITER.md`.
- FACT: `04_Nuclear-Power-and-Three-Stage-Programme.md` - essential fission-versus-fusion distinction.

- FACT: `10_National-Quantum-Mission-and-Quantum-Tech.md` - high-end research infrastructure, computation and advanced science linkages.
- FACT: `16_Nanotechnology-and-Applications.md` - superconducting magnets, materials stress and high-technology manufacturing relevance.
- FACT: `20_Emerging-Materials-Rare-Earths-and-Critical-Minerals.md` - long-term energy strategy and technology-sovereignty context.

Core answer architecture - fusion, ITER and long-horizon energy

Thesis choice. Fusion is a scientific and industrial-learning frontier, not an operating source of grid electricity; a good answer distinguishes plasma gain from a power plant.

10-mark spine. Define D-T fusion and confinement; identify ITER as an experimental tokamak; name India's domestic/ITER contribution; close with the status and implication in one qualified line.

15/20-mark spine. Structure as **fusion mechanism and three readiness thresholds -> India's research/industrial contribution -> possible energy and technology gains -> tritium, materials, cost and schedule limits**.

Evidence units.

- **Claim:** D-T fusion needs an integrated physical condition, not temperature alone -> **Lawson triple product and magnetic confinement in a tokamak** -> explains why plasma stability is the central challenge -> **qualification:** a laboratory shot or a Q claim does not include plant electricity, magnets, cryogenics and conversion losses.
- **Claim:** India participates as an industrial and scientific contributor -> **ITER-India within IPR delivers in-kind packages including the cryostat, shielding, cooling, cryogenic, RF, neutral-beam, power and diagnostic systems** -> participation builds high-end manufacturing, procurement and plasma expertise -> **qualification:** supplying components does not mean India owns an operational fusion reactor.
- **Claim:** ITER's success would be conditional long-term value -> **ITER targets Q at an experimental facility; IPR's Aditya-U and SST-1 remain research platforms** -> the programme can advance materials, control and energy science -> **qualification:** tritium breeding, neutron damage, divertor heat loads, regulation and commercial economics remain unsolved at power-plant scale.

Verdict. State the global-energy promise as contingent on later DEMO-class engineering, not as a near-term substitute for fission, renewables or efficiency.

CLOSING RECALL FLOW - CORE SYNTHESIS - ITER SCHEDULE DEMO COMMERCIAL LADDER AND DATED AUDIT

```
START / CONCEPT: CORE SYNTHESIS - ITER schedule DEMO commercial ladder and dated audit
        |
        v
EXACT TERMS: CORE SYNTHESIS · ITER schedule DEMO commercial ladder · dated audit · Mechanism chain
· Qualified use · Subject
        |
        v
MECHANISM / ARGUMENT: Mechanism chain: DEMO-commercial boundary - Volatile fusion boundary
Qualified use: Close with scheduled targets, DEMO prerequisites and dated evidence.
        |
        v
CONSEQUENCE / CONTRAST: ITER is designed as an experimental bridge between present fusion machines
and later electricity-generating demonstration plants; it is not itself a commercial power plant.
        |
        v
UPSC TRAP / ANSWER-USE: FACT: Fusion is different from fission: fusion combines light nuclei, while
fission splits heavy nuclei. FACT: A tokamak is a magnetic-confinement device; ITER follows the
magnetic-confinement route, not inertial confinement. FACT: ITER is an experimental project
intended to demonstrate feasibility and bridge the gap toward later power-producing systems; it is
not a commercial plant. FACT: India is one of the ITER members and contributes components,
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research stage. FACT: No answer should claim that an operational fusion power plant already exists
anywhere in the world. FACT: ITER's design target is Q ≥ 10 - 500 MW of fusion power from 50 MW of
injected heating power.
        |
        v
ANSWER-GRABBING FORMULATION: Read ITER schedule DEMO commercial ladder and dated audit as a
sequence: identify DEMO-commercial boundary, follow the named mechanism to its user or output, and
stop at the last verified status rather than assuming the next stage.
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SCIENCE AND TECHNOLOGY DEEP-REVIEW CORE CONTROL

- **Must remember:** Fusion joins light nuclei in a high-temperature plasma; confinement, heating, stability, plasma gain, blanket, tritium and materials problems separate a physics experiment from a power plant.
- **Close distinction:** Plasma Q is not engineering breakeven, tokamak is not stellarator, magnetic is not inertial confinement, ITER is not a commercial generator, and an experimental pulse is not continuous net-electricity production.
- **Mechanism / status / evidence limit:** State quantity, device, pulse/energy boundary, participant role, date and experimental status; keep ITER, DEMO concepts and commercial deployment on separate maturity rungs.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Fusion-fission boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: A. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q2. Which option preserves the technical boundary of Fusion-fission boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: B. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q3. Which statement uses Fusion-fission boundary without changing its institution, unit or status?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: C. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q4. Which option avoids the standard UPSC close-option trap about Fusion-fission boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: D. Explanation: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q5. Which statement correctly identifies DT-reaction boundary?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: A. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q6. Which option preserves the technical boundary of DT-reaction boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: B. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q7. Which statement uses DT-reaction boundary without changing its institution, unit or status?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: C. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q8. Which option avoids the standard UPSC close-option trap about DT-reaction boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: D. Explanation: The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q9. Which statement correctly identifies Temperature-plasma boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: A. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q10. Which option preserves the technical boundary of Temperature-plasma boundary?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: B. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q11. Which statement uses Temperature-plasma boundary without changing its institution, unit or status?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: C. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q12. Which option avoids the standard UPSC close-option trap about Temperature-plasma boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: D. Explanation: Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q13. Which statement correctly identifies Lawson boundary?

A. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: A. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other

options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q14. Which option preserves the technical boundary of Lawson boundary?

A. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: B. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q15. Which statement uses Lawson boundary without changing its institution, unit or status?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: C. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q16. Which option avoids the standard UPSC close-option trap about Lawson boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.

Answer: D. Explanation: The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q17. Which statement correctly identifies Magnetic-inertial boundary?

A. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: A. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q18. Which option preserves the technical boundary of Magnetic-inertial boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: B. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q19. Which statement uses Magnetic-inertial boundary without changing its institution, unit or status?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: C. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q20. Which option avoids the standard UPSC close-option trap about Magnetic-inertial boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: D. Explanation: Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q21. Which statement correctly identifies Tokamak-stellarator boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: A. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q22. Which option preserves the technical boundary of Tokamak-stellarator boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: B. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q23. Which statement uses Tokamak-stellarator boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: C. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q24. Which option avoids the standard UPSC close-option trap about Tokamak-stellarator boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.

Answer: D. Explanation: A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q25. Which statement correctly identifies Q-plasma boundary?

A. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: A. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q26. Which option preserves the technical boundary of Q-plasma boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: B. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q27. Which statement uses Q-plasma boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: C. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q28. Which option avoids the standard UPSC close-option trap about Q-plasma boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

Answer: D. Explanation: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q29. Which statement correctly identifies Engineering-breakeven boundary?

A. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: A. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q30. Which option preserves the technical boundary of Engineering-breakeven boundary?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: B. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q31. Which statement uses Engineering-breakeven boundary without changing its institution, unit or status?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: C. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q32. Which option avoids the standard UPSC close-option trap about Engineering-breakeven boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.

Answer: D. Explanation: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q33. Which statement correctly identifies Alpha-neutron boundary?

A. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: A. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q34. Which option preserves the technical boundary of Alpha-neutron boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: B. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q35. Which statement uses Alpha-neutron boundary without changing its institution, unit or status?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: C. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q36. Which option avoids the standard UPSC close-option trap about Alpha-neutron boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.

Answer: D. Explanation: Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q37. Which statement correctly identifies Tritium-breeding boundary?

A. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: A. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q38. Which option preserves the technical boundary of Tritium-breeding boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: B. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q39. Which statement uses Tritium-breeding boundary without changing its institution, unit or status?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: C. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement,

mission rung or operating status.

Q40. Which option avoids the standard UPSC close-option trap about Tritium-breeding boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.

Answer: D. Explanation: Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q41. Which statement correctly identifies Materials boundary?

A. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: A. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q42. Which option preserves the technical boundary of Materials boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: B. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q43. Which statement uses Materials boundary without changing its institution, unit or status?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: C. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q44. Which option avoids the standard UPSC close-option trap about Materials boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.

Answer: D. Explanation: High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q45. Which statement correctly identifies Divertor boundary?

A. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: A. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q46. Which option preserves the technical boundary of Divertor boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: B. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q47. Which statement uses Divertor boundary without changing its institution, unit or status?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: C. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the

scientific category, institutional owner, measurement, mission rung or operating status.

Q48. Which option avoids the standard UPSC close-option trap about Divertor boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Answer: D. Explanation: A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q49. Which statement correctly identifies ITER-purpose boundary?

A. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: A. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q50. Which option preserves the technical boundary of ITER-purpose boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: B. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q51. Which statement uses ITER-purpose boundary without changing its institution, unit or status?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: C. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating

status.

Q52. Which option avoids the standard UPSC close-option trap about ITER-purpose boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.

Answer: D. Explanation: ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q53. Which statement correctly identifies ITER-Q boundary?

A. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: A. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q54. Which option preserves the technical boundary of ITER-Q boundary?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: B. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q55. Which statement uses ITER-Q boundary without changing its institution, unit or status?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: C. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q56. Which option avoids the standard UPSC close-option trap about ITER-Q boundary?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Answer: D. Explanation: ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q57. Which statement correctly identifies Membership-contribution boundary?

A. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: A. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q58. Which option preserves the technical boundary of Membership-contribution boundary?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: B. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q59. Which statement uses Membership-contribution boundary without changing its institution, unit or status?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: C. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission

rung or operating status.

Q60. Which option avoids the standard UPSC close-option trap about Membership-contribution boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.

Answer: D. Explanation: ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q61. Which statement correctly identifies IPR-ITERIndia boundary?

A. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: A. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q62. Which option preserves the technical boundary of IPR-ITERIndia boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: B. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q63. Which statement uses IPR-ITERIndia boundary without changing its institution, unit or status?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: C. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q64. Which option avoids the standard UPSC close-option trap about IPR-ITERIndia boundary?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. D. IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

Answer: D. Explanation: IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q65. Which statement correctly identifies Domestic-device boundary?

A. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: A. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q66. Which option preserves the technical boundary of Domestic-device boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: B. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q67. Which statement uses Domestic-device boundary without changing its institution, unit or status?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: C. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q68. Which option avoids the standard UPSC close-option trap about Domestic-device boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

Answer: D. Explanation: ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q69. Which statement correctly identifies Schedule-status boundary?

A. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. B. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: A. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q70. Which option preserves the technical boundary of Schedule-status boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: B. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q71. Which statement uses Schedule-status boundary without changing its institution, unit or status?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: C. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q72. Which option avoids the standard UPSC close-option trap about Schedule-status boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

Answer: D. Explanation: ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q73. Which statement correctly identifies DEMO-commercial boundary?

A. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: A. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q74. Which option preserves the technical boundary of DEMO-commercial boundary?

A. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. B. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. C. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: B. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q75. Which statement uses DEMO-commercial boundary without changing its institution, unit or status?

A. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. D. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.

Answer: C. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q76. Which option avoids the standard UPSC close-option trap about DEMO-commercial boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Answer: D. Explanation: A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q77. Which statement correctly identifies Volatile fusion boundary?

A. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. B. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. C. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. D. Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.

Answer: A. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q78. Which option preserves the technical boundary of Volatile fusion boundary?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. C. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. D. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.

Answer: B. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain

plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q79. Which statement uses Volatile fusion boundary without changing its institution, unit or status?

A. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. D. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.

Answer: C. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

Q80. Which option avoids the standard UPSC close-option trap about Volatile fusion boundary?

A. A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. B. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. C. Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. D. Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Answer: D. Explanation: Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The other options change the scientific category, institutional owner, measurement, mission rung or operating status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

Audited ledgers route the direct 2025 GS-III ITER contribution and future-energy demand here. No other direct PYQ or answer key is manufactured.

PYQ DEMAND CARD 1 - 2025 GS-III

Demand: Mention India's contributions to ITER and explain the implications of project success for global energy.

Status: Verified direct routed Mains demand.

Model solution: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50

MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Demand decoding: The directive **answer** requires a direct position on “PYQ DEMAND CARD 1 - 2025 GS-III”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Analytical body:

1. **Claim and named evidence:** Demand: Mention India's contributions to ITER and explain the implications of project success for global energy. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Fusion-fission boundary: Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental. **Q-plasma**

boundary: Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “PYQ DEMAND CARD 1 - 2025 GS-III”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.
- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named**

evidence/example: Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain why ITER must not be described as a commercial power plant. Answer in about 150 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.

- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **answer** requires a direct position on “Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty,

exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Distinguish plasma Q from engineering and electricity breakeven. Answer in about 150 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250 words.

Model thesis: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.
- Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.
- The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.
- Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Qualified conclusion: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **explain** requires a direct position on “Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: DT-reaction boundary. **Named evidence/example:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Alpha-neutron boundary. **Named evidence/example:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Explain the D-T fusion chain and its tritium and materials constraints. Answer in about 250…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.

Model thesis: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.
- IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.

- ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.
- Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Qualified conclusion: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **assess** requires a direct position on “Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are

different claims. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

2. **Claim and named evidence:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Domestic-device boundary. **Named evidence/example:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Assess India's domestic fusion ecosystem and ITER participation. Answer in about 250 words.”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer in about 300 words.

Model thesis: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.
- The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.
- Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.
- A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.
- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.
- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.

Qualified conclusion: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Demand decoding: The directive **evaluate** requires a direct position on “Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Analytical body:

1. **Claim and named evidence:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
2. **Claim and named evidence:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
3. **Claim and named evidence:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
4. **Claim and named evidence:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
5. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
6. **Claim and named evidence:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
7. **Claim and named evidence:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** Connect mechanism -> named system/institution -> application or

implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

8. **Claim and named evidence:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
9. **Claim and named evidence:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
10. **Claim and named evidence:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Temperature-plasma boundary. **Named evidence/example:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Lawson boundary. **Named evidence/example:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Magnetic-inertial boundary. **Named evidence/example:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tokamak-stellarator boundary. **Named evidence/example:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Divertor boundary. **Named evidence/example:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Evaluate fusion through scientific, engineering and commercial feasibility thresholds. Answer…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Analyse the implications of ITER success without overstating timelines or electricity output. Answer in about 300 words.

Model thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Materials boundary. **Named evidence/example:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-purpose boundary. **Named evidence/example:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** ITER-Q boundary. **Named evidence/example:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Membership-contribution boundary. **Named evidence/example:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** IPR-ITERIndia boundary. **Named evidence/example:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Schedule-status boundary. **Named evidence/example:** ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** DEMO-commercial boundary. **Named evidence/example:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Volatile fusion boundary. **Named evidence/example:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary.

Claim -> named evidence -> analysis -> qualification:

- Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
- Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
- Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
- High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
- ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
- ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.
- ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.
- IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.
- ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.

- A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.
- Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Demand decoding: The directive **analyse** requires a direct position on “Analyse the implications of ITER success without overstating timelines or electricity output.…”, each clause, the exact technical mechanism, named Indian institution, dated status, application, risk and qualification.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Analytical body:

1. **Claim and named evidence:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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9. **Claim and named evidence:** ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.
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11. **Claim and named evidence:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries. **Analysis:** Connect mechanism -> named system/institution -> application or implementation pathway -> public or strategic consequence. **Qualification:** State platform, unit, source/date/status, maturity, regulatory limit, uncertainty, exception or residual risk.

Counter-position / limit: A launch, test, target, budget, capacity, approval, contract, dataset, benchmark or prototype cannot alone establish deployment, induction, commercial operation, safety, inclusion or observed outcome; test mechanism, status, scale and evidence.

Qualified conclusion: Claim: Q-plasma boundary. **Named evidence/example:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Engineering-breakeven boundary. **Named evidence/example:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer. **Analysis:** This fixes the scientific process, system boundary, institution, application and verified capability rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, unit, orbit, signal, mission, reactor, fuel-cycle or experimental-status boundary. **Claim:** Tritium-breeding boundary. **Named evidence/example:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; 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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism -> system/institution -> application/status -> risk/outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for unit, date, actor, platform, approval/test/deployment rung and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains science rather than listing schemes, uses India-centric evidence and preserves technical, institutional, regulatory and maturity distinctions.

How to improve this answer: For “Analyse the implications of ITER success without overstating timelines or electricity output.…”, replace the weakest catalogue point with one exact mechanism, named mission/institution/platform, dated status, measurable boundary, implementation constraint and answer-specific qualification.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

> **Subject:** Science & Technology | **Tier:** Advanced | **GS Paper:** GS-III + Prelims. **Core area:** Experimental fusion strategy, international collaboration, India’s domestic research base and realism about commercial timelines. **Grounded in:** ITER fusion pages (https://www.iter.org/sci/whatisfusion ; https://www.iter.org/fusion-energy/what-will-iter-do ; https://www.iter.org/fusion-energy/after-iter); ITER India contributions/update pages (https://www.iter.org/node/20687/indias-contributions-iter ; https://www.iter.org/node/20687/updated-baseline-presented ; https://www.iter.org/sites/default/files/media/2025-06/ic-36_press_release_final2.pdf); IPR SST-1 and Aditya-U pages (https://nsd.ipr.res.in/documents/sst-1.html ; https://nsd.ipr.res.in/documents/aditya_u.html); official IPR ecosystem references (https://www.ipr.res.in/); ITER new-baseline release, 8 Jul 2024 (https://www.iter.org/node/20687/new-baseline-prioritize-robust-start-exploitation); ITER mission overview (https://www.iter.org/few-lines); ITER-India contributions page (https://www.iterindia.in/indias-contribution-iter); IPR ADITYA-U and SST-1 pages (https://www.ipr.res.in/rdactivities/Aditya-U1 ; https://nsd.ipr.res.in/documents/sst-1.html) - re-verified 2 Aug 2026. FACT: = source-grounded | ANALYSIS: = analytical inference | CURRENT: = current/dated development. *Companion:* `basic/05_Nuclear-Fusion-and-ITER.md`.

1. Analytical frame

ANALYSIS: Fusion answers fail when they treat "unlimited clean energy" as an imminent policy option. The disciplined frame separates three distinct thresholds: **scientific feasibility** (can a plasma release more fusion energy than is injected into it - the physics gain Q), **engineering feasibility** (can a machine run continuously, breed its own tritium, survive 14 MeV neutron damage and produce net *electrical* output after subtracting the plant's own consumption), and **commercial viability** (capital cost, availability factor, licensing regime, supply chain for superconductors, beryllium/tungsten and lithium). ITER is designed to demonstrate the first and to test components for the second; it is not designed to generate electricity at all. India's analytical significance therefore lies in *capability capture* - cryogenics, superconducting magnets, radio-frequency heating sources, vacuum vessels and remote handling built by Indian industry for ITER are dual-use industrial competences that pay off regardless of when fusion power arrives.

2. Visual foundation

```
FUSION TECHNOLOGY LADDER
plasma physics -> confinement -> heating/current drive -> materials resilience
-> tritium handling/breeding -> long-duration operation -> future DEMO-class plants

KEY REALITY
ITER = experimental feasibility platform
DEMO-like future systems = later power-generation ambition
```

Advanced distinction	Why it matters
Fusion vs fission	Prevents present-power and future-research confusion
Tokamak vs power plant	A device can demonstrate physics without generating grid electricity
Experimental milestone vs commercial readiness	Guards against hype-driven answers
India in ITER vs India at home	International participation and domestic capability both matter

3. Essential definitions

Concept	Exam-ready meaning
FACT: **Burning plasma ambition**	Fusion state where plasma heating is substantially sustained by fusion reactions themselves - physically, when the **3.5 MeV alpha particles** retained in the plasma supply most of the heating. This is the frontier no tokamak has yet crossed at scale.
ANALYSIS: **Lawson triple product (n·T·τ_E)**	The quantitative statement of what "hard" means in fusion: density, temperature and energy-confinement time must be raised *simultaneously*. Machine size, magnetic field strength and plasma stability all enter through τ_E.
ANALYSIS: **Q, engineering Q and net electricity**	**Q** = fusion power / injected heating power (ITER target ≥ 10). **Engineering Q** subtracts the whole plant's recirculating consumption - magnets, cryoplant, pumps, heating efficiency, thermal-to-electric conversion. A machine can achieve Q > 1 and still be a net electricity *consumer*; conflating the two is the commonest fusion error in exam writing.
ANALYSIS: **Tritium breeding ratio (TBR)**	Tritium bred per tritium consumed. **TBR must exceed 1** (with margin for decay and losses) or a D-T plant cannot fuel itself, since world tritium inventories are tiny. Test blanket modules on ITER exist precisely to validate this.
ANALYSIS: **Neutron-resistant materials**	14.1 MeV neutrons cause displacement damage, helium/hydrogen production and activation in structural steel. Reduced-activation ferritic-martensitic steels and tungsten armour are the candidate answers; validating them needs a dedicated high-flux neutron source, which is itself an unbuilt prerequisite for DEMO.
FACT: **DEMO**	Future demonstration-class fusion power concept expected after ITER-type experimentation, not before it - the first machine intended to put fusion electricity on a grid.
FACT: **Cryogenic / superconducting ecosystem**	Engineering base needed for large magnetic-confinement devices; also the clearest channel through which fusion spending yields non-fusion industrial returns (MRI magnets, accelerators, cryogenics, vacuum technology).
FACT: **Fusion hype risk**	Analytical error of mistaking experimental success for immediate commercial power deployment.
FACT: **Long-duration plasma operation**	Ability to sustain controlled plasma for longer pulses, central to future practicality. A tokamak's induced plasma current is inherently pulsed, which is why *steady-state* operation (SST-1's purpose) and stellarators are strategically interesting.
FACT: **In-kind contribution model**	International collaboration structure in which members provide components and systems rather than only cash support - attractive because it converts a contribution into **domestic industrial capability**, but demanding because delivery failures by one member delay everyone.

4. Mechanism / how it works

1. Fusion requires extreme conditions of temperature and confinement because light nuclei naturally repel one another.
2. Magnetic-confinement systems like tokamaks try to hold plasma stable long enough to sustain useful fusion conditions.
3. Experimental fusion progress depends on simultaneous advances in magnets, materials, diagnostics, plasma control, vacuum systems and fuel handling.

4. ITER is designed to demonstrate scientific and technological feasibility, but not to deliver commercial electricity to the grid.
5. Therefore, the real pathway is staged: research machines -> ITER-class feasibility -> later DEMO-like systems -> only then possible commercial fusion plants.

5. Institutions and programmes

Core institutions

- FACT: **ITER Organization:** global flagship experimental fusion collaboration of **seven members - India, EU, China, Japan, South Korea, Russia and the USA** - sited at Saint-Paul-lez-Durance, France.
- FACT: **India as ITER member:** contributes **nine in-kind packages** - cryostat, in-wall shielding, cooling-water system, cryogenic system, ion-cyclotron RF heating, electron-cyclotron RF heating, diagnostic neutral beam, power supplies and diagnostics.
- FACT: **Institute for Plasma Research (IPR), Gandhinagar:** a DAE-aided institute; India's plasma-physics and fusion-research base, operating ADITYA-U and SST-1.
- FACT: **ITER-India:** the **Indian Domestic Agency for ITER, constituted as a dedicated project within IPR**. It contracts Indian industry, manages delivery schedules and interfaces with the ITER Organization. ANALYSIS: Writing "IPR / ITER-India" as one entity blurs a research institute with a procurement agency - precisely the kind of institutional conflation UPSC penalises.
- FACT: **SST-1 and Aditya-U:** domestic experimental platforms that cultivate operational knowledge and scientific manpower; IPR describes them as research devices, not power reactors.

Governance / strategic debates

- ANALYSIS: Fusion cooperation is simultaneously scientific diplomacy, industrial capability-building and long-horizon energy strategy.
- ANALYSIS: Because fusion is still experimental, policymakers must balance visionary funding with realistic public communication.
- ANALYSIS: International collaboration accelerates learning but also requires domestic industry depth to convert participation into durable technological gain.
- ANALYSIS: A serious fusion policy discussion asks what India learns in magnets, cryogenics, control systems and materials - not only whether it gets future electricity.
- ANALYSIS: **The in-kind model's double edge:** it guarantees Indian firms build first-of-a-kind hardware to nuclear-grade tolerances (real capability transfer), but it also couples India's programme to a multi-party schedule it does not control - the 2024 re-baselining is the practical illustration.
- ANALYSIS: **Regulatory vacuum:** fusion devices are not fission reactors, yet they involve tritium inventories, activated materials and large stored magnetic energy. India has no distinct fusion-specific licensing framework; whether fusion should be regulated under the atomic-energy law or a lighter bespoke regime is an open governance question worth one line in a Mains answer.
- ANALYSIS: **Private fusion is now a real variable globally**, with venture-funded compact-tokamak and alternative-confinement ventures claiming faster timelines than ITER. The analytically honest position is that none has demonstrated a burning plasma or a closed tritium cycle, so private entry changes the *funding landscape* more than the *technical readiness*.

6. Indian applications, strategic significance and limitations

Strategic and economic significance

- FACT: Fusion R&D strengthens India’s high-technology base in superconducting systems, vacuum engineering, advanced materials and precision manufacturing.
- FACT: ITER participation gives Indian industry exposure to a first-of-its-kind global engineering project.
- FACT: Domestic tokamaks prevent fusion expertise from being only externally dependent.
- ANALYSIS: Even before commercial fusion, capability in these areas can spill over into strategic industry and research ecosystems.

Implementation constraints

- ANALYSIS: Plasma stability, neutron damage, tritium-cycle complexity and cost remain major barriers.
- ANALYSIS: Large experimental projects have long schedules and are vulnerable to repair, baseline and integration delays.
- ANALYSIS: Fusion's economics remain uncertain even if physics milestones improve.
- ANALYSIS: Public narratives that imply “cheap unlimited power soon” are analytically unsound.

Safety / ethics / governance caution

- ANALYSIS: Fusion is safer in some respects than fission, but it is not a zero-governance technology; materials activation, tritium handling and complex engineering still matter.
- ANALYSIS: Policymakers must communicate uncertainty honestly to avoid eroding public trust through exaggerated claims.

7. Must-Know Facts for Prelims

- FACT: Fusion combines light nuclei; fission splits heavy nuclei.
- FACT: ITER is an experimental fusion facility and is not intended to be an operational commercial power plant.
- FACT: India is one of the ITER members and participates through component and technology contributions.
- FACT: SST-1 and Aditya-U are Indian tokamak platforms tied to domestic fusion research.
- FACT: Magnetic confinement and inertial confinement are distinct approaches.
- FACT: No operational commercial fusion power plant exists anywhere at present.

8. UPSC traps

- WRONG: ITER will soon solve India's electricity shortage. -> ITER is a scientific/technological feasibility project, not a near-term electricity solution.
- WRONG: Fusion eliminates the need to study fission. -> Today's nuclear power reality remains fission-based.
- WRONG: Any successful plasma milestone means commercial deployment is close. -> Physics success does not remove materials, fuel-cycle and economic barriers.
- WRONG: India's role in ITER is symbolic only. -> It also builds industrial and technological capability through contributions.
- WRONG: Fusion has no safety or regulatory dimension. -> Tritium handling, activated materials and large-scale engineering still require governance.

9. CURRENT: Current anchor

- CURRENT: **08 Jul 2024 | ITER new baseline - adopted.** ITER moved from a "First Plasma" milestone to **Start of Research Operations in 2034**, **full magnetic energy in 2036** and **deuterium-tritium operation in 2039**, retaining the mission target of **Q ≥ 10 (500 MW fusion from 50 MW injected heating)**.
- CURRENT: **17 Feb 2025 | India's ITER contribution - highlighted officially.** ITER emphasised India's cryostat and broader contribution footprint across its nine in-kind packages.
- CURRENT: **19 Jun 2025 | ITER Council Meeting 36 - strong project execution.** Official press release welcomed sector-module installation ahead of schedule and completion of all major magnets.
- CURRENT: **Verified position, Aug 2026 | Domestic platforms.** IPR describes **ADITYA-U** (deuterium plasma, divertor and shaped-plasma campaigns) and **SST-1** (steady-state superconducting tokamak) as experimental research devices. **Status: research operation, not power generation.**
- CURRENT: **Ongoing official position:** after ITER, later demonstration plants remain the next step; fusion therefore stays in the experimental/research category.

Verified current anchor	Analytical use in answers
2024 re-baselining to Start of Research Operations 2034 and D-T 2039	Use to show that frontier science projects are governed by schedule realism, not linear optimism - and to date any "implications of success" answer to the second half of the century rather than the current decade.
Q ≥ 10 stated as **500 MW thermal from 50 MW injected heating**	Use to make the precise point that ITER's target is a *plasma physics* gain, not net electricity; ITER has no generating equipment at all.
India's nine in-kind packages, cryostat highlighted in 2025	Use to argue that ITER participation has industrial and diplomatic value beyond physics alone: Indian firms delivered nuclear-grade cryogenics, vacuum and RF systems that are reusable in accelerators, space and medical technology.
ITER Council welcomed 2025 construction milestones	Use to show progress without falsely claiming commercial readiness.
ADITYA-U and SST-1 described as research devices	Use to keep India's domestic claim honest: India has plasma-physics capability and steady-state ambition, not a fusion power programme.
"After ITER" framing remains central	Use to reinforce that no operational fusion power plant yet exists anywhere, and that DEMO-class machines plus a validated tritium-breeding blanket and a neutron-materials test facility all remain prerequisites.

ANALYSIS: **Currentness note:** The dated statuses above are accurate to the cited source date (latest re-verification 2 Aug 2026); verify later updates before exam use.

11. Mains framework / angles

- ANALYSIS: Begin by separating fusion from fission clearly and immediately.
- ANALYSIS: Move from plasma-confinement challenge to international collaboration to India's domestic capability base.
- ANALYSIS: Add a realistic policy frame: strategic technology learning now, uncertain electricity benefits later.
- ANALYSIS: Conclude that fusion matters today mainly as science, engineering and industrial capability-building rather than as an immediate grid solution.

> **Answer thesis:** Fusion should be analysed as a frontier research ecosystem rather than as a present energy source: ITER matters because it tests feasibility, India's participation matters because it builds industrial and scientific depth, and domestic tokamak work matters because technological sovereignty in advanced science is built through sustained experimentation - yet none of this justifies claiming that commercial fusion power already exists or is immediately deployable.

12. Probable questions

- ANALYSIS: **Prelims (practice):** Which one of the following correctly distinguishes ITER, DEMO-type future plants and present-day fission reactors?
- ANALYSIS: **Mains (10 marks, practice):** Why is it misleading to present ITER as an operational fusion power plant? Answer in 150 words.
- ANALYSIS: **Mains (15 marks, practice):** Evaluate the scientific, industrial and strategic significance of India's participation in ITER while explaining why fusion still remains a long-term rather than immediate power option.

13. Study links

- FACT: Foundation companion: `basic/05_Nuclear-Fusion-and-ITER.md`.
- FACT: `04_Nuclear-Power-and-Three-Stage-Programme.md` - essential contrast with fission reality.
- FACT: `10_National-Quantum-Mission-and-Quantum-Tech.md` - high-end research infrastructure and computation linkages.

- FACT: `16_Nanotechnology-and-Applications.md` - superconductors, materials stress and enabling technologies.
- FACT: `20_Emerging-Materials-Rare-Earths-and-Critical-Minerals.md` - long-horizon energy and strategic-technology context.

CONSOLIDATED REGISTER NOTES

Nuclear Fusion and ITER: SYSTEM, MISSION AND INSTITUTION MAP

1. **Fusion-fission boundary:** Nuclear fusion combines light nuclei, while nuclear fission splits heavy nuclei; commercial nuclear electricity today is fission-based and fusion remains experimental.
2. **DT-reaction boundary:** The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating.
3. **Temperature-plasma boundary:** Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness.
4. **Lawson boundary:** The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition.
5. **Magnetic-inertial boundary:** Magnetic confinement holds a lower-density plasma with magnetic fields, while inertial confinement rapidly compresses a small fuel target; the two approaches use different time and density regimes.
6. **Tokamak-stellarator boundary:** A tokamak is a toroidal magnetic-confinement device using a strong plasma current, while a stellarator relies on externally shaped coils for steady confinement; neither term means a commercial power plant.
7. **Q-plasma boundary:** Fusion gain Q compares fusion power produced in the plasma with external heating power injected into the plasma; Q above one is not plant-wide net electricity.
8. **Engineering-breakeven boundary:** Engineering or electricity breakeven must include magnets, cryogenics, pumps, heating efficiency and thermal conversion; a machine can have favourable plasma Q while remaining a net electricity consumer.
9. **Alpha-neutron boundary:** Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures.
10. **Tritium-breeding boundary:** Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle.
11. **Materials boundary:** High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement.
12. **Divertor boundary:** A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.
13. **ITER-purpose boundary:** ITER is an experimental tokamak under construction to study burning-plasma conditions and integrated fusion technologies; it has no turbine and is not a commercial electricity plant.
14. **ITER-Q boundary:** ITER's official design objective is 500 MW of fusion power from 50 MW of injected plasma heating, Q=10; the official page explicitly says this heat will not be converted to electricity.
15. **Membership-contribution boundary:** ITER has seven members, while India's participation is delivered substantially through specified in-kind procurement packages; membership share, cash cost and component delivery are different claims.
16. **IPR-ITERIndia boundary:** IPR is a DAE-aided plasma-research institute, while ITER-India is the domestic agency organised as a project within IPR for procurement and delivery; research institute and project agency are not synonyms.
17. **Domestic-device boundary:** ADITYA-U and SST-1 are Indian experimental tokamaks for plasma and long-pulse research; neither is a power reactor or evidence of commercial fusion generation.

18. **Schedule-status boundary:** ITER's re-baselined research, magnetic-energy and deuterium-tritium milestones are future project targets; construction progress does not convert them into completed operations.
19. **DEMO-commercial boundary:** A later DEMO-class system is intended to bridge experimentation toward electricity production, while commercial deployment additionally needs availability, licensing, fuel self-sufficiency and acceptable cost.
20. **Volatile fusion boundary:** Temperature, gain, pulse duration, schedule, cost, member contribution, component status and milestone claims require dated ITER, IPR or ITER-India evidence and must retain plasma-versus-plant boundaries.

Nuclear Fusion and ITER: CATEGORY, STATUS AND NUMBER FIREWALLS

- Do not merge fusion with fission power.
- Do not quote temperature as proof of energy gain.
- Do not omit confinement time from fusion conditions.
- Do not merge magnetic and inertial confinement.
- Do not call every magnetic device a tokamak.
- Do not call a tokamak a power plant.
- Do not equate plasma Q with net electricity.
- Do not omit plant recirculating power.
- Do not treat alpha particles and neutrons identically.
- Do not assume tritium is freely available.
- Do not call fusion radiation-free or materials-free.
- Do not omit divertor and exhaust constraints.
- Do not call ITER a commercial reactor.
- Do not rewrite 500 MW fusion heat as electricity.
- Do not invent member shares, cost or contribution values.
- Do not merge IPR with ITER-India.
- Do not call ADITYA-U or SST-1 power reactors.
- Do not turn scheduled ITER milestones into completed status.
- Do not merge ITER, DEMO and commercial plants.
- Do not invent current gain, duration, schedule or cost claims.

Nuclear Fusion and ITER: ANSWER-WRITING SPINE

```
SEPARATE FUSION FROM FISSION AND RESEARCH FROM POWER DEPLOYMENT
-> DEFINE D-T REACTION PLASMA LAWSON CONFINEMENT AND DEVICE TYPE
-> SEPARATE PLASMA Q ENGINEERING BREAKEVEN AND NET ELECTRICITY
-> TRACE ALPHA HEATING NEUTRON BLANKET TRITIUM MATERIALS AND DIVERTOR
-> MAP ITER INDIA IPR ITER-INDIA ADITYA-U SST-1 AND DEMO
-> DATE-STAMP RESEARCH CONSTRUCTION SCHEDULE CONTRIBUTION AND MILESTONE
-> CONCLUDE ON LONG-HORIZON ENERGY PLUS PRESENT INDUSTRIAL LEARNING
```

Nuclear Fusion and ITER: LIVE-SOURCE AND VOLATILE-CLAIM BOUNDARY

Substantive ITER text fetched on 2026-09-03 confirms the experimental Q=10 objective, no-electricity boundary, technology-integration purpose and tritium-breeding test role. Schedule, cost and package-status claims remain date-bounded.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Nuclear-process firewall

```
FISSION -> splits heavy nuclei and powers current reactors
FUSION -> combines light nuclei in experimental systems
COMMERCIAL STATUS -> fission operating, fusion not deployed
SHARED WORD NUCLEAR -> does not erase process differences
RULE -> Topic 05 cannot substitute for Topic 04
```

ASCII MASTER FLOW - PANEL 2/12: D-T energy map

```
DEUTERIUM + TRITIUM -> fusion reaction
HELIUM-4 / ALPHA -> charged product retained for self-heating
NEUTRON -> carries most energy out of plasma
BLANKET -> captures heat and supports fuel functions
RULE -> particle role controls engineering consequence
```

ASCII MASTER FLOW - PANEL 3/12: Fusion-condition triangle

```
TEMPERATURE -> raises probability of overcoming repulsion
DENSITY -> supplies reacting particles
CONFINEMENT TIME -> retains energy long enough
LAWSON TRIPLE PRODUCT -> all three must combine
TRAP -> a temperature record alone proves little
```

ASCII MASTER FLOW - PANEL 4/12: Confinement comparison

```
MAGNETIC -> lower density and longer confinement
TOKAMAK -> torus with strong plasma current
STELLARATOR -> externally shaped steady-field geometry
INERTIAL -> dense target and extremely short confinement
RULE -> approach and device name are not power status
```

ASCII MASTER FLOW - PANEL 5/12: Breakeven ladder

```
PLASMA Q -> fusion power divided by injected heating
Q ABOVE ONE -> plasma-level scientific threshold
ENGINEERING Q -> includes recirculating plant loads
NET ELECTRICITY -> includes thermal conversion and all consumption
COMMERCIALITY -> adds availability, cost and licensing
```

ASCII MASTER FLOW - PANEL 6/12: Neutron-to-material chain

```
14.1 MeV NEUTRON -> escapes magnetic confinement
BLANKET -> receives energy and may breed tritium
STRUCTURE -> displacement damage and transmutation
ACTIVATION -> radioactive material-management issue
LIFETIME -> separate engineering feasibility constraint
```

ASCII MASTER FLOW - PANEL 7/12: Tritium fuel loop

```
D-T PLASMA -> consumes tritium
FUSION NEUTRON -> enters lithium-bearing blanket
LITHIUM REACTION -> produces new tritium
EXTRACTION / PROCESSING -> returns usable fuel
RULE -> a test blanket is not a closed commercial cycle
```

ASCII MASTER FLOW - PANEL 8/12: Plasma exhaust path

```
CORE PLASMA -> creates heat and helium ash
EDGE TRANSPORT -> moves particles and heat outward
DIVERTOR -> exhausts impurities and helium
PLASMA-FACING MATERIAL -> receives extreme heat load
RULE -> confinement success does not solve exhaust lifetime
```

ASCII MASTER FLOW - PANEL 9/12: ITER mission card

```
UNDER CONSTRUCTION -> experimental international tokamak
BURNING-PLASMA STUDY -> core scientific purpose
Q=10 DESIGN -> 500 MW fusion from 50 MW plasma heating
NO TURBINE -> no conversion to grid electricity
TEST TECHNOLOGIES -> bridge to later machines
```

ASCII MASTER FLOW - PANEL 10/12: India-in-ITER map

```
INDIA -> one of seven ITER members
IN-KIND PACKAGES -> specified hardware and systems
IPR -> DAE-aided plasma-research institute
ITER-INDIA -> domestic procurement and delivery agency in IPR
RULE -> membership, institute, agency and package differ
MUST REMEMBER: Fusion joins light nuclei in a high-temperature plasma; confinement,...
```

ASCII MASTER FLOW - PANEL 11/12: Domestic research devices

```
ADITYA-U -> experimental tokamak and plasma campaigns
SST-1 -> superconducting long-pulse research platform
RESEARCH OUTPUT -> diagnostics, control and operating knowledge
NOT OUTPUT -> commercial fusion electricity
RULE -> experimental operation is not power generation
CLOSE DISTINCTION: Plasma Q is not engineering breakeven, tokamak is not stellarator,...
```

ASCII MASTER FLOW - PANEL 12/12: Fusion-readiness answer spine

```
DEFINE -> D-T fusion plasma confinement Lawson and Q
TRACE -> alpha heating neutron blanket tritium and materials
SEPARATE -> plasma gain engineering gain and net electricity
MAP -> ITER India IPR ITER-India ADITYA-U SST-1 and DEMO
VERIFY -> dated schedule cost contribution and milestone evidence
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State quantity, device, pulse/energy...
```